



Behavioral Sequence Analysis of Mouse Visual Navigation in Virtual Reality

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Introduction

- Virtual reality is a robust way to study neural activity in mice
- Understanding their motor behavior on such set up is crucial to correlate it to neural activity
- Mice behavior can be characterized into stereotyped syllables [1]
- Strategic movement strings of syllables for successful task completions creates unique sequences

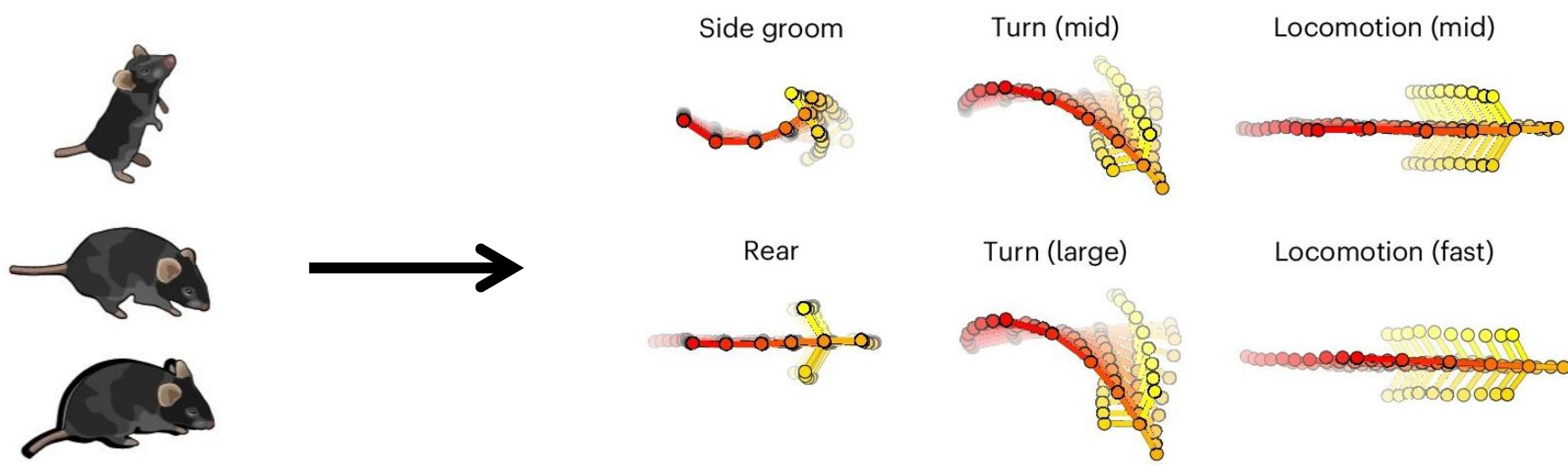


Figure 1: Physical mouse behavior to moseq syllable sequence

Study Goals

- Characterize expert mice motor control strategies on VR task set up
- Examine pose and motor differences between expert and naive mice

Methods

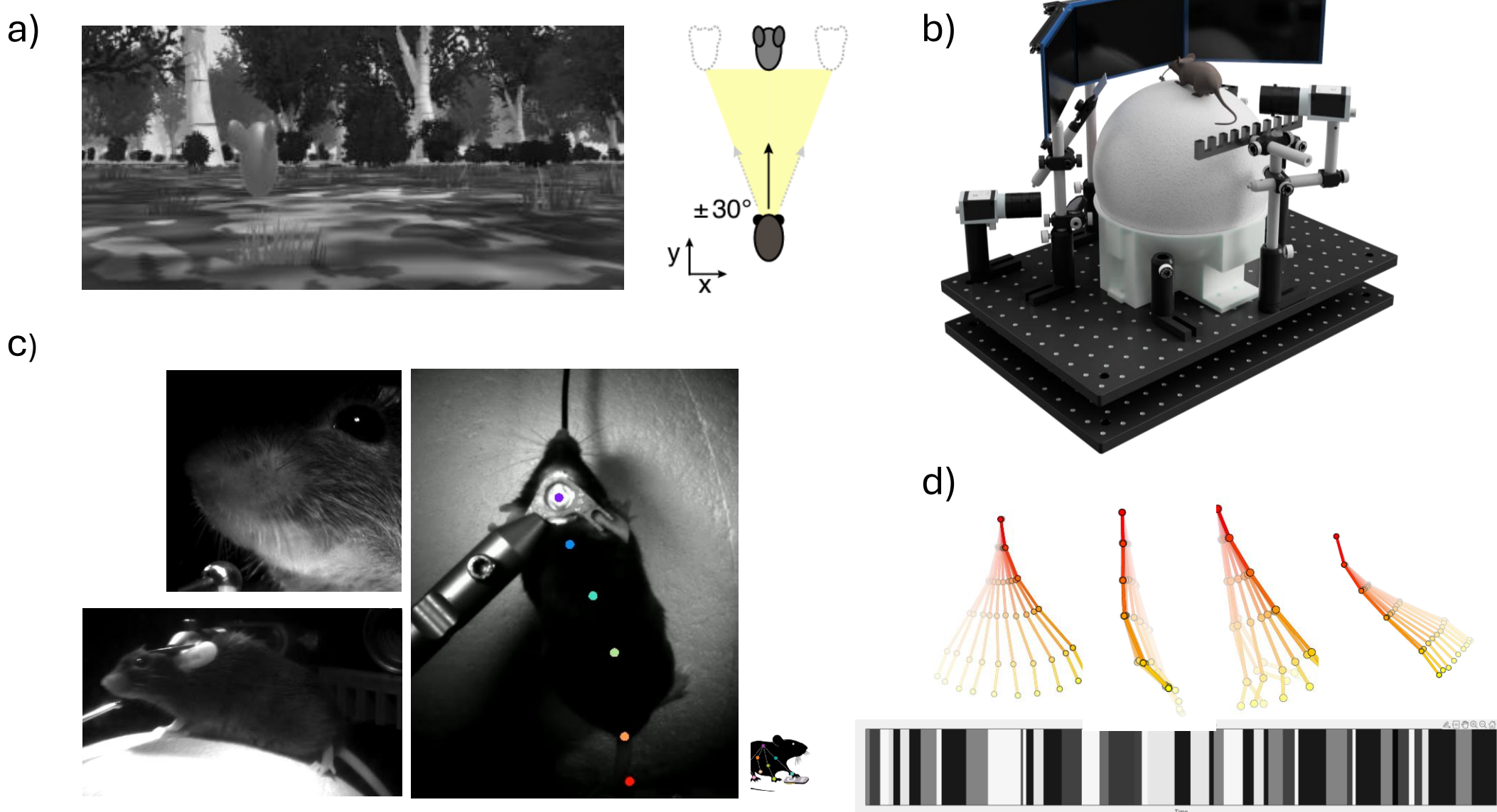


Figure 2: Data Collection
a. Virtual task b. VR set up rig with cameras for data collection c. Data collection angles from the top, pose, and eye. Top camera data processed through DeepLabCut [2] d. Keypoint Moseq syllable outputs with behavioral sequence

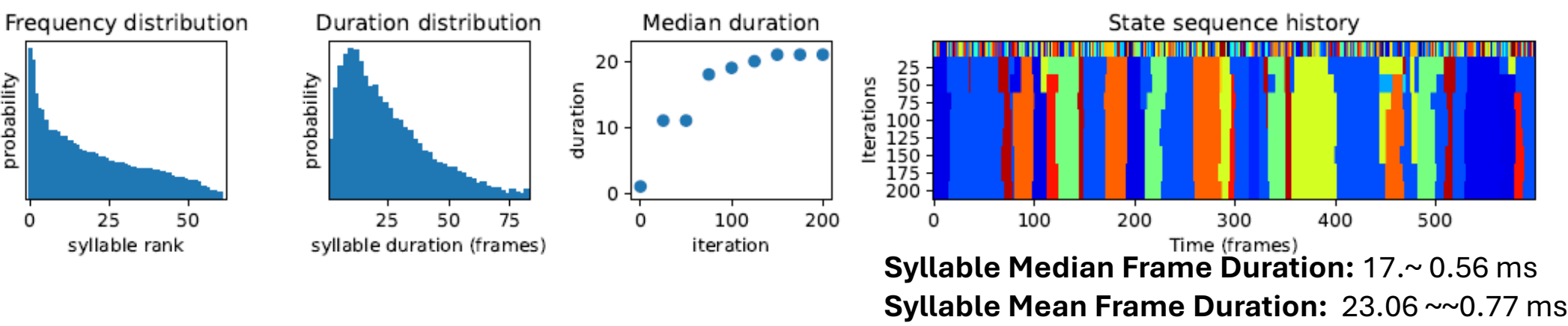


Figure 3: Moseq Model Fitting
A total of 14 expert sessions and 4 naive recording sessions from 6 different mice used to train Keypoint Moseq model.

Results

Syllables Organized by Direction and Activity

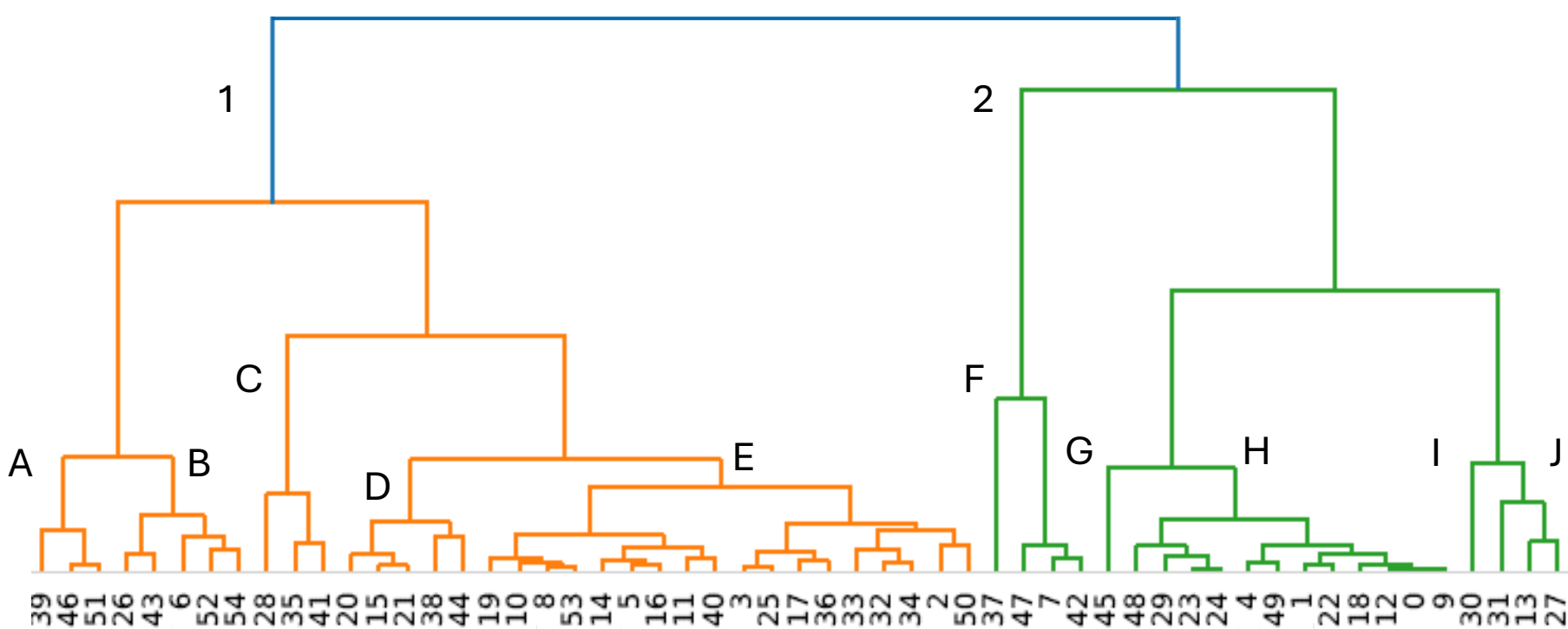


Figure 3: Syllables Fitted from Mice VR are Organized by Direction and Activity
1: Left Syllables 2: Right Syllables
A: L-Swing B: L-Pause and Push Maneuver C: L-Tail whip Turns D: Gradual Direction Change E: L-Active Direction Change F: R-swing G: R-Active Turn H: R-Push Turn I: R-Slow turn with tail J: Turn Scrunch

Syllable Analysis of Expert Behavior

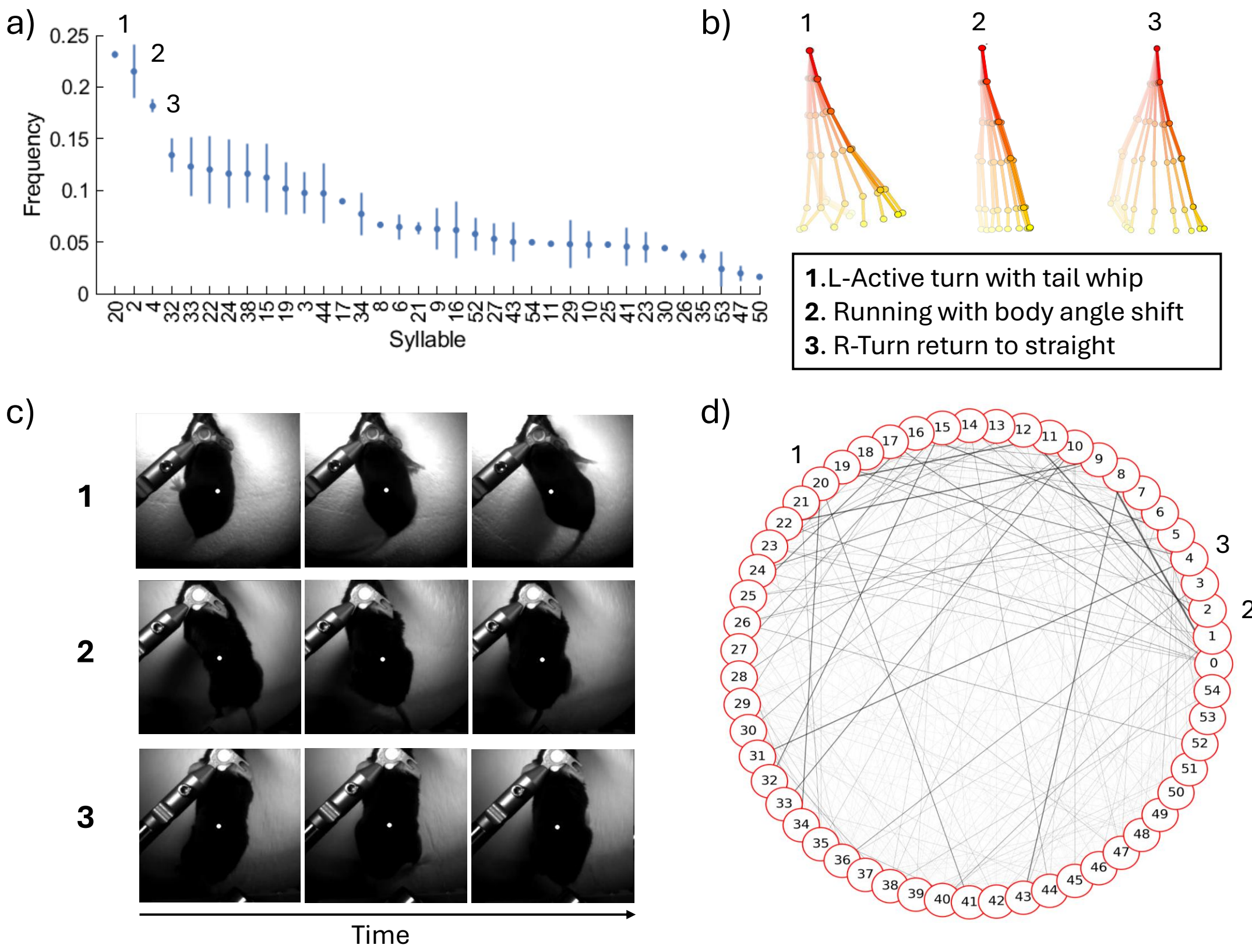


Figure 4: Expert Mice Employ Consistent Motor Patterns
A. Syllable Frequency with top 3 most frequent syllables labeled 1, 2, and 3 respectively B. Trajectory plot of syllable 1, 2, and 3 C. Examples of syllable 1, 2, and 3 D. Transitions between syllables

Expert vs Naive Comparison

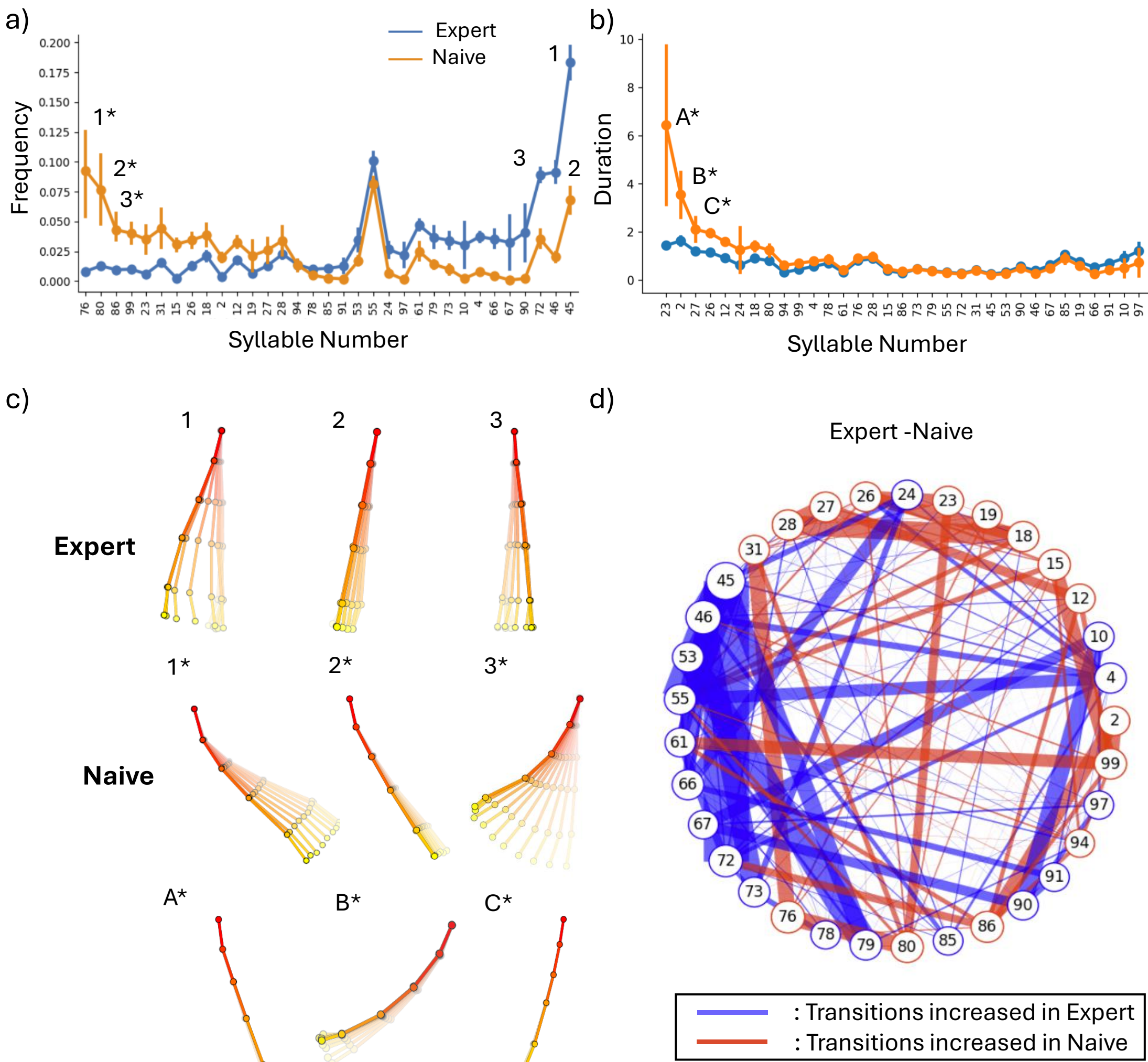


Figure 5: Expert and Naive Mice Exhibit Distinct Motor Patterns
A. Expert vs Naive Syllable Frequency Differences, Most expert dominant labeled 1, 2, 3 and most naive labeled 1*, 2*, 3* B. Duration Spent in Syllable, naive longest duration labeled A*, B*, and C* C. Trajectory plots of labeled syllable breakdown D. Expert vs Naive Transition Weight

Conclusions

- Mouse behavior on a VR ball set up is describable by Moseq
- Expert Mice use consistent sequences enriched with controlled tail-supported turning
- Naive behavior includes large swings starting from extreme body angles, and extended duration at extreme angles

References

- Weinreb, C., Pearl, J.E., Lin, S. *et al.* Keypoint-MoSeq: parsing behavior by linking point tracking to pose dynamics. *Nat Methods* **21**, 1329–1339 (2024). <https://doi.org/10.1038/s41592-024-02318-2>
- Nath, T., Mathis, A., Chen, A.C. *et al.* Using DeepLabCut for 3D markerless pose estimation across species and behaviors. *Nat Protoc* **14**, 2152–2176 (2019). <https://doi.org/10.1038/s41596-019-0176-0>